From Plant-Level Traditional Use to Compound-Level Target Activity: Leakage-Controlled Multiple-Instance Learning for Vietnamese Medicinal Plants

Minh Nhat
Corresponding author: minhnhatpro2504@gmail.com

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Abstract

Background: Traditional-medicine databases connect plants to reported uses and plants to constituent metabolites, whereas modern bioactivity databases measure activity at the compound–target level. Naively assigning every plant-level use to every reported metabolite creates false labels and obscures which chemical evidence, if any, supports the use. We study whether Vietnamese ethnopharmacological annotations can improve molecular-target ranking without making this cross-granularity assumption.

Methods: We specify a confirmatory workflow linking a maintainer-authorized, versioned VietHerb snapshot to ChEMBL 37 activities for human DPP4, acetylcholinesterase, and COX-2. Each plant is represented as a bag of standardized metabolites, with gated-attention multiple-instance learning and positive–unlabelled treatment of absent use annotations. Before restricted VietHerb data were available, we validated the activity branch on frozen ChEMBL 37 using a 0.70 Tanimoto connected-component split, five seeds, separate and shared fingerprint MLPs, and dependence-aware paired bootstrap intervals.

Results: The frozen ChEMBL 37 extraction contained 43,843 candidate records and 16,941 eligible records, yielding 10,698 labelled compound–target pairs and 10,686 unique molecules after replicate, contradiction, and gray-zone handling. No molecule, InChIKey, Murcko scaffold, or Tanimoto-connected component crossed train, validation, and test partitions; maximum test-to-train Tanimoto was 0.698 under the locked 0.70 threshold. Separate target MLPs achieved macro PR-AUC 0.9281 ± 0.0035, versus 0.9203 ± 0.0024 for shared activity-only learning. The paired shared-minus-separate difference was -0.0077 (hierarchical bootstrap 95% CI, -0.0141 to -0.0021), although the shared model improved BEDROC20. These results reveal metric- and target-dependent negative transfer but do not test the ethnopharmacological MIL hypothesis.

Conclusions: The implemented workflow provides a leakage-audited basis for testing whether plant-level knowledge adds compound-level ranking value. Frozen ChEMBL 37 results validate the activity branch and motivate explicit negative-transfer controls, while the central MIL comparison remains prospectively locked pending authorized VietHerb data and expert-audited therapeutic-use mappings.

1 Introduction

Natural products remain an important source of chemical diversity for drug discovery, but their biological evidence is heterogeneous and frequently recorded at a different level from the molecule being screened. VietHerb was created to organize Vietnamese medicinal plants together with metabolites, therapeutic effects, morphology, and geography. Its original release reported 2,881 species, 10,887 metabolites, and many-to-many species–metabolite and species–effect relations [Nguyen-Vo et al., 2019]. This local resource is scientifically valuable, but a reported use of a plant is neither evidence that every constituent is active nor proof of clinical efficacy.

Prior computational ethnopharmacology already links traditional symptoms, efficacies, herbs, ingredients, and molecular targets. HTINet and HTINet2 use heterogeneous networks and knowledge graphs for herb–target prediction [Wang et al., 2019, Duan et al., 2024]; other systems use clinical symptom profiles or multimodal ingredient–target information [Cheung, 2026, Liang et al., 2025]. Consequently, the broad claim that traditional knowledge has not been used for target prediction is untenable. The unresolved issue addressed here is narrower: VietHerb annotations are plant-level labels, while ChEMBL observations are compound-level measurements. This is naturally a multiple-instance learning (MIL) problem, in which a plant is a bag and its reported metabolites are instances [Ilse et al., 2018, Doran and Ray, 2012].

Reliable evaluation is equally important. Ligand benchmarks can reward memorization when close analogues cross partitions [Wallach and Heifets, 2018], and conventional scaffold splitting does not eliminate every form of chemical redundancy. Traditional-medicine data add botanical leakage because the same metabolite can occur in several species. Missing compound–target and plant–use edges also cannot automatically be interpreted as negatives, making positive–unlabelled (PU) learning a relevant sensitivity analysis [Kiryo et al., 2017, Wang et al., 2026].

We therefore make four contributions. First, we define an auditable data model that preserves compound, plant, therapeutic-concept, assay, and provenance relations. Second, we formulate cross-granularity supervision with attention MIL instead of label broadcasting. Third, we introduce split manifests that keep identical structures, related chemistry, and plant components disjoint. Fourth, we isolate the value of ethnopharmacological information with matched architectures and falsification controls. The preregistered primary question is whether plant-bag supervision improves compound-level out-of-distribution PR-AUC beyond activity-only multi-task learning.

2 Materials and methods

2.1 Data sources and governance

VietHerb provides plant–metabolite and plant–therapeutic-effect relations [Nguyen-Vo et al., 2019]. Because no explicit database redistribution licence was found during our audit, the confirmatory analysis requires a maintainer-authorized export. Raw rows will not be redistributed without written permission. Published artifacts will otherwise contain code, schemas, source identifiers, checksums, aggregate counts, and split-generation logic. Traditional-use records are treated as hypothesis-generating cultural and historical evidence, not as validated biomedical claims.

For the confirmatory study, measured activities were extracted from the versioned ChEMBL 37 SQLite release; the archive, database, and output were bound by SHA-256 hashes. The earlier live REST snapshot is retained only as an exploratory engineering analysis. ChEMBL is a manually curated, large-scale bioactivity resource with explicit assay and document provenance [Zdrazil et al., 2024]. LOTUS may be used only for an optional occurrence/provenance sensitivity analysis [Rutz et al., 2022]. Aggregated natural-product databases are not assumed to provide independent activity ground truth [Chandrasekhar et al., 2025].

2.2 Target-activity cohort

The pilot targets are human dipeptidyl peptidase 4 (DPP4, CHEMBL284), acetylcholinesterase (AChE, CHEMBL220), and prostaglandin G/H synthase 2 (COX-2/PTGS2, CHEMBL230). The primary cohort contains high-confidence, single-protein biochemical binding assays with standardized IC_{50} or K_i values in nM. Functional assays form a separate sensitivity cohort. Records with data-validity warnings, duplicate flags, unresolved structures, or ambiguous target assignments are excluded. Exact standardized structure–target replicates are aggregated by median pChEMBL; pairs spanning at least one log unit are flagged as contradictory and excluded from the primary analysis.

The prespecified binary endpoint labels pChEMBL ≥ 6 as active and pChEMBL ≤ 5 as inactive; the interval $(5, 6)$ is excluded as a gray zone. Only exact (equality) measurements enter the primary binary cohort. Alternative cutoffs and continuous pActivity models are sensitivity analyses. Unmeasured compound–target pairs remain masked. All stereochemistry and structure standardization decisions, rejected rows, release metadata, and SHA-256 hashes are retained in a machine-readable manifest.

2.3 Therapeutic-use normalization

Original text is preserved. Candidate mappings are retrieved from frozen MeSH/Mondo releases, after which a language model may rank candidates but may not accept a label automatically. The stored record includes source span, ontology version, candidate list, confidence, model/prompt hash, abstention, and reviewer decisions. Two pharmacognosy or pharmacy reviewers independently annotate a prespecified stratified sample before adjudication. We report exact and ancestor-aware precision, recall, F1, raw agreement, and Cohen's κ; sample size is selected for a target confidence-interval width rather than fixed solely by convenience.

The first model uses three coarse, clinically interpretable families: metabolic/diabetes, cognition/neurological use, and pain/inflammation. ATC is not used as the primary disease ontology because it classifies active drug substances rather than diseases or symptoms.

2.4 Molecular and botanical split construction

Molecules are standardized globally and assigned once across all targets and plants. Two locked regimes answer different questions. The chemical-OOD regime groups molecules by a prespecified fingerprint clustering rule and reports train–test nearest-neighbour Tanimoto distributions. A scaffold component split is retained as a reproducible secondary regime. The botanical cold-start regime holds out plant taxa and explicitly handles metabolites shared between taxa.

For the strict joint split, molecule nodes are joined when they share a chemical group or occur in the same plant. Entire transitive connected components are assigned to train, validation, or test. Thus, if compound A is chemically linked to B and B occurs with C in one species, all three remain in one partition. No feature transformation, loss weight, class prior, or hyperparameter is fitted using test data.

2.5 Multiple-instance multi-task model

Let x_{ij} denote metabolite j reported in plant i, and let $h_{ij} = f_\theta(x_{ij})$ be its shared molecular representation. The primary heads predict measured compound–target activities with masked, class-weighted binary cross entropy. For each plant bag, gated attention computes

$$e_{ij} = w^\top \left[\tanh(V h_{ij}) \odot \sigma(U h_{ij})\right], \tag{1}$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}, \tag{2}$$

$$z_i = \sum_j a_{ij} h_{ij}, \tag{3}$$

followed by one plant-use head per therapeutic family [Ilse et al., 2018]. Attention weights provide instance attribution but are not interpreted as causal explanations.

An observed plant use is positive and absence is unlabelled. Auxiliary heads therefore use non-negative PU risk [Kiryo et al., 2017]. For class prior π, positive risk R_p^+, positive-as-negative risk R_p^-, and unlabelled negative risk R_u^-, the unbiased estimate is

$$R_{\mathrm{PU}} = \pi R_p^+ + R_u^- - \pi R_p^-, \tag{4}$$

with the non-negative correction applied when the empirical negative component falls below zero. Priors are frozen from training-only estimation and varied in sensitivity analyses. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{activity}} + \lambda \mathcal{L}_{\text{plant-MIL}}. \tag{5}$$

The prespecified compute-feasible confirmatory encoder is a radius-2, 2,048-bit chiral Morgan fingerprint followed by a two-layer MLP. This deliberately separates the scientific contribution from an expensive foundation model. A D-MPNN/Chemprop baseline [Yang et al., 2019] and one pretrained molecular encoder are secondary comparisons using identical split IDs.

2.6 Baselines and falsification controls

Prespecified models are: per-target fingerprint logistic regression; random forest or gradient boosting; separate neural target models; shared activity-only multi-task learning; naive plant-label broadcasting; the proposed MIL model; and optional domain importance weighting. Controls shuffle use labels among training plants while preserving prevalence, replace uses with equally sized random auxiliary labels, compare expert-verified with confidence-weighted mappings, and equalize parameter counts and update counts. The primary comparison is MIL versus the architecture-matched activity-only model. Naive broadcasting is an ablation, not the proposed method.

2.7 Training and statistical analysis

The MLP uses hidden dimensions 1,024 and 512, dropout 0.25, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}, batch size 256, and early stopping patience 12 on validation macro PR-AUC. Auxiliary weights $\{0.1, 0.3, 1.0\}$ are selected on validation data. Seeds are 13, 37, 71, 101, and 137.

The confirmatory primary endpoint is the paired change in macro test PR-AUC. We additionally report target-wise PR-AUC and prevalence, BEDROC ($\alpha = 20$), EF1% and EF5%, precision/recall at fixed experimental budgets, AUROC, Brier score, and calibration. PR-AUC is emphasized for imbalanced screening [Saito and Rehmsmeier, 2015], while BEDROC measures early recognition [Truchon and Bayly, 2007]. Paired 95% intervals use 1,000 bootstrap replicates with structure cluster as the resampling unit for compound endpoints and plant as the unit for bag endpoints. Target-level secondary tests use Benjamini–Hochberg correction.

For the frozen activity-only comparison, target-specific MLPs and an architecture-matched shared MLP were trained with five prespecified seeds. Within each run, early stopping used validation macro PR-AUC and the test set was evaluated only after restoring the best validation checkpoint. The source, dataset, run, metric, and prediction artifacts are bound by SHA-256. Runs used Python 3.11.9, RDKit 2025.09.5, and PyTorch 2.5.1 with CUDA 12.1 on an NVIDIA RTX 4060 Laptop GPU with 8 GB memory.

3 Results

3.1 Cohort and shift audit

The frozen ChEMBL 37 database yielded 43,843 candidate records for the three targets. Prespecified assay, measurement, relation, validity, duplicate, and structure filters retained 16,941 rows. After standardization and replicate aggregation, 435 contradictory pairs and 3,353 gray-zone pairs were excluded, leaving 10,698 labelled structure–target pairs and 10,686 unique molecules. Chemical-component assignment produced 7,480 training, 1,603 validation, and 1,603 test molecules spanning 4,417 Murcko scaffolds. Pairwise overlap between every split was zero for molecule identifier, InChIKey, scaffold, and connected component. Test-to-train nearest-neighbour Tanimoto

had median 0.500, 95th percentile 0.673, and maximum 0.698, below the locked 0.70 clustering threshold. Table 1 summarizes the target observations. No VietHerb row was used.

Table 1: Frozen ChEMBL 37 activity cohort after curation. Active denotes pChEMBL ≥ 6; inactive denotes pChEMBL ≤ 5. This table validates the confirmatory activity branch but contains no VietHerb supervision.

Endpoint	Measurements	Molecules	Active (%)	Test positives
DPP4	3,725	3,725	93.4	522
AChE	4,026	4,026	71.3	431
COX-2	2,947	2,947	77.8	344
All targets	10,698	10,686	80.8	1,297
Plant-use bags	0	0	—	0

The absence of identity, scaffold, and fingerprint-neighbour overlap satisfies the implemented chemical leakage gates. Bag-size distributions, plants per metabolite, and botanical component sizes cannot be estimated until the authorized VietHerb snapshot is frozen; these remain locked diagnostics.

3.2 Frozen ChEMBL 37 activity-only comparison

Separate target MLPs exceeded the shared activity-only MLP on the primary macro PR-AUC endpoint (Table 2). The mean paired shared-minus-separate difference was -0.0077. A hierarchical bootstrap resampling model seed and chemical component gave a 95% interval of -0.0141 to -0.0021. Target-wise PR-AUC differences were $+0.0045$ for DPP4, -0.0272 for AChE, and -0.0005 for COX-2, localizing most negative transfer to AChE. The shared model nevertheless improved BEDROC20 and EF1%, so its earliest-ranked candidates were competitive despite worse global precision–recall and calibration. This comparison measures activity-only parameter sharing and is not evidence for auxiliary plant supervision.

Table 2: Frozen ChEMBL 37 chemical-OOD test performance, reported as mean ± standard deviation across five seeds. EF1% is an enrichment factor, not a percentage.

Model	Macro PR-AUC	BEDROC20	EF1%	Brier
Separate target MLPs	0.9281 ± 0.0035	0.9714 ± 0.0066	1.2334 ± 0.0245	0.1833 ± 0.0132
Shared activity MLP	0.9203 ± 0.0024	0.9803 ± 0.0036	1.2527 ± 0.0000	0.2253 ± 0.0263

3.3 Implementation and confirmatory readiness

All 39 unit tests passed, including masked-loss, nnPU-risk, attention-MIL, metric, and connected-component leakage invariants. A CUDA smoke run exercised the activity-only, naive-broadcast, MIL, and shuffled-MIL paths end to end on deterministic synthetic bags. Its manifest is explicitly marked `scientific_results=false`; therefore synthetic performance is not reported. Botanical cold starts, PU-prior and auxiliary-weight sensitivity, and attention stability require real bags and remain locked confirmatory analyses.

4 Discussion

The frozen activity results establish that the version-pinned curation, chemical-OOD splitting, training, and evaluation stack works on measured bioactivity. They also demonstrate negative transfer: hard sharing reduced the primary macro PR-AUC, driven mainly by AChE, even while

improving early-ranking metrics. This metric dependence argues against declaring one architecture superior from AUROC or enrichment alone and supports the prespecified reporting of ranking, calibration, and target-wise outcomes.

The future principal result must be interpreted as incremental predictive value under the locked evaluation, not validation of traditional medicine. If MIL improves over activity-only learning but not over shuffled labels, the gain is consistent with generic regularization rather than semantic ethnopharmacological signal. If naive broadcasting performs similarly to MIL, chemical or botanical leakage and label granularity require further investigation. If all ethnomedical variants underperform, negative transfer is an informative result and should not be hidden by selecting favorable targets.

Docking or short molecular dynamics may provide a later structure-based plausibility assessment, but neither constitutes biological validation. Docking scores have well-known limitations for affinity ranking [Warren et al., 2006]. Prospective biochemical assays with positive, negative, interference, and purity controls are required to claim target inhibition.

Limitations include incomplete or changing plant databases, uncertain traditional-use normalization, incomplete chemical inventories for each plant, assay heterogeneity, class-prior uncertainty in PU learning, and the possibility that attention is not a faithful explanation. Written data permission and a domain collaborator are release gates rather than administrative afterthoughts.

5 Conclusion

We present a reproducible, leakage-controlled formulation for learning from plant-level traditional-use labels and compound-level target measurements without broadcasting each use label to every constituent. A version-pinned ChEMBL 37 activity benchmark of 10,686 compounds passed identity, scaffold, and fingerprint-neighbour split audits. Shared activity-only learning reduced macro PR-AUC relative to separate target MLPs, providing a concrete negative-transfer baseline for the later auxiliary experiment. The central claim—that ethnopharmacological bag supervision improves out-of-distribution molecular prioritization—is not yet supported or refuted. It will be tested only after an authorized VietHerb export and expert-audited mappings pass the remaining release gates.

Data and code availability

Code, configurations, schemas, tests, source locks, and derived manifests are available at `https://github.com/MinhNhat-2504/vietherb-mtl`. ChEMBL-derived artifacts follow CC BY-SA 3.0. VietHerb row-level data will be released only if the maintainers provide explicit permission; otherwise the repository will publish acquisition instructions, hashes, identifiers, and reproducible transformation code without redistributing restricted rows.

Author contributions, funding, and competing interests

Minh Nhat: conceptualization, methodology, software, investigation, data curation, validation, visualization, and writing. No external funding was used for the reported exploratory computation. The author declares no competing interests. Contributions and declarations must be updated if domain reviewers or experimental collaborators join the confirmatory study.

Ethics statement

The reported work used public chemical bioactivity records and deterministic synthetic bags for software validation; it involved no human participants, animals, or clinical intervention. Traditional-use annotations are not treated as clinical evidence.

Generative-AI assistance disclosure

A generative-AI coding assistant was used for implementation support, test generation, literature-search assistance, and language editing. The author reviewed the source code, citations, numerical outputs, and manuscript claims and remains responsible for the final work.

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